Code No:134BU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, February -2024
OPERATING SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Define an operating system.	[2]
b)	What services do operating systems provide?	[3]
c)	Define process synchronization.	[2]
d)	What is the purpose of monitors in synchronization?	[3]
e)	Define the term “thrashing”.	[2]
f)	How is the allocation of frames determined in a virtual memory environment?	[3]
g)	Define the term "free-space management".	[2]
h)	What is the role of directory implementation in storage management?	[3]
i)	What is a deadlock in a system?	[2]
j)	Write the three conditions that characterize a deadlock.	[3]

PART – B

(50 Marks)

2.a) Compare the user view and the system view of an operating system.
b) Explore the computer system organization, providing examples to illustrate their roles in system functionality. [5+5]

OR

3.a) Explain different computing environments and how they impact operating system design and functionality.
b) Discuss about memory management in operating systems. [6+4]

4.a) Describe the functions of the Process Control Block in process management.
b) Explore the operations on processes in Unix, focusing on fork(), exec(), wait(), and exit(). [5+5]

OR

5. Compare and contrast different scheduling criteria used in process scheduling. [10]

6.a) Describe the various types of memory management strategies.
b) Explain the concept IA-32 paging. [6+4]

OR

7.a) Analyze the copy-on-write mechanism and its advantages in optimizing memory operations.
b) Differentiate between virtual addresses and physical addresses. [6+4]

8.a) Explain how does file sharing enhance collaboration and resource utilization in a file system.

b) Describe different types of allocation methods. [5+5]

OR

9.a) Discuss the efficiency and performance considerations in file system design and implementation.

b) Explain the components and functions of disk structure in storage management. [5+5]

10.a) Explain the principles of protection.

b) Discuss recovery strategies from deadlock. How can a system recover from a deadlock situation? [4+6]

OR

11. Briefly explain the following terms.
a) Capability-based systems.
b) System Protection. [5+5]

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